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Stochastic optimization model for the weekly scheduling of a hydropower system in day-ahead and secondary regulation reserve markets

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ABSTRACT

Hydropower stands out for its fast response ability and flexible operation, playing a predominant role in the provision of regulation reserves. As hydro power is an energy-constrained generation technology, it needs to be protected against any possible deployment of the scheduled reserves. However, the few models that formulate a detailed hourly co-optimization of energy and regulation reserves, neglect infeasibilities that could be derived from the requested reserves in real-time. As the regulation reserve market is becoming increasingly important, hydro producers can no longer neglect such effect. This paper presents a stochastic optimization model to find the optimal hourly schedule of a set of hydraulically coupled hydropower plants to obtain the weekly operation that simultaneously maximizes the expected profit in both energy and regulation reserves markets. The model is formulated for a price-taker agent, and it considers a very detailed representation of the system including minimum–maximum water flows, net head dependency, and fractional water travel time. The main contribution is that the obtained solution protects a multi-reservoir system against risk of water and storage unavailability due to the uncertainty in real-time use of regulation-up and down reserves, respectively, and that the reserves deem the net head dependency. The paper presents a realistic case study where the proposed formulation has been tested successfully with real data from the Spanish electricity market.

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1. Introduction

The increasing penetration level of renewable energy sources in many power systems is changing the pattern of energy and ancillary services prices in electricity markets [1]. In this framework, hydro generation companies need to update their models and scheduling tools in order to maximize their expected profit taking advantage of the price spread that can be observed between the energy and reserve prices in real markets (especially in periods with a substantial share of wind power in the generation mix).

Hydropower units are more flexible than low variable cost technologies such as nuclear and coal thermal plants, as they can change their generation from the minimum to maximum capacity in just few seconds. In addition to this, hydropower generation involves lower variable costs comparing to flexible thermal technologies such as open gas cycle units. Therefore, the hydropower technology plays an important role in the provision of the secondary regulation service in many systems all around the world.

The management of a hydrosystem in both the long-term and short-term is not an easy task. A hydropower producer must decide the hourly operation of reservoirs in order to optimize its participation in competitive electricity markets. In the literature, short-term hydropower scheduling problem has been treated widely. However, only some few works have formulated a co-optimization model from the perspective of a hydro producer who tries to maximize the expected profit in both the day-ahead electricity market and the secondary regulation reserve market.

In [2], a deterministic short-term model (24 hourly periods) for a hydro system participating in day-ahead electricity market, spinning reserve market and regulation market is described. With similar features than [2], the model presented in [3] introduces stochasticity in market prices, treating water inflows as a deterministic input data. It is important to highlight that spinning reserves (10S) and non-spinning reserves (10N) calculated by [3] are not

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Nomenclature

Some sets, subsets, parameters and variables are also shown graphically in Fig. 1.

Sets and indexes

G, g set and index of hydropower units. One hydropower plant can be made of several units (turbines)
J, j set and index of water arcs
K, k set and index of discrete generation curves for each net head considered in the approximation
N, n set and index of nodes of the hydro system: reservoirs (e), hydropower plants (c) and natural water inflows (i)
P, p set and index of time periods, running from 1 to P
W, w set and index of uncertainty scenarios
N', P', W' alias of previously defined sets

Subsets

$BG(g)$ set of hydropower units allowed to participate in the secondary regulation reserver market
$C(n)$ sets of nodes which are hydropower plant, formed by at least one hydropower unit
$CG(n)$ unit or set of units which belong to node $n \in C(n)$
$Dq(n)$ set of arcs of water discharged flow which begin in node n
$Ds(n)$ set of arcs of water spilled flow which begin in node n
$E(n)$ set of nodes which are reservoirs
$EG(g)$ reservoir associated to hydropower unit g
$I(n)$ set of nodes representing water inflow locations which are not placed in a reservoir
$PG(g)$ set of hydropower units which are pumped storage units
$UPC(n)$ reservoir or set of reservoirs from which the units in node $n \in C(n)$ receive water flow directly
$Uq(n)$ set of arcs of water discharged flow which finish in node n
$Us(n)$ set of arcs of water spilled flow which finish in node n

Parameters

$\delta^{t}_{w,k,g,p}$ generating energy coefficient in curve k of unit g in period p and scenario w (MW/h m^3/h)
$\lambda EM_{w,p}$, $\lambda SM_{w,p}$ day-ahead electricity market price (€/MWh) and secondary regulation reserve market price (€/MW) in period p and scenario w
$\tau^{q/s}_{p',p,j}$ percentage of water discharged/spilled flow from upstream of arc j in period p' reaching downstream of arc j in period p due to time delays
$\mathcal{F}^{up}$ maximum percentage of the offered secondary regulation-up reserve requested by TSO in real time
$\mathcal{F}^{dw}$ maximum percentage of the offered secondary regulation-down reserve requested by TSO in real time
l_p time lenght of period p (h)
M a big number, 10^6
$NI_{w,n,p}$ natural inflow (or outflow in case it takes a negative value) at node n, in scenario w and period p (h m^3/h)
OP_g maximum start-up and shutdown maneuvers per day in hydropower unit g
$\bar{p}_g, \bar{q}_g$ maximum technical generation (MW) and water flow (h m^3/h) of unit g
$\bar{p}^{b}_{g}, \bar{q}^{b}_{g}$ maximum technical consumption (MW) and pumped water flow (h m^3/h) of unit $g \in PG$
$\bar{p}_{w,k,g,p}, \underline{p}_{w,k,g,p}$ maximum and minimum hydropower generation in curve k of unit g in period p and scenario w (MW)
$\bar{q}_n, \underline{q}_n$ maximum and minimum total water flow through node n due to design characteristics (h m^3/h)
$\bar{q}_{w,k,g,p}, \underline{q}_{w,k,g,p}$ maximum and minimum water flow in curve k of unit g in period p and scenario w (h m^3/h)
$\bar{rf}_{n,p}$, $\underline{rf}_{n,p}$ maximum and minimum water discharged and/or spilled flow obligations imposed by water authorities in node n which is a reservoir or a run-of-the-river hydropower plant (h m^3/h)
TD_g minimum down time (h)
TU_g minimum up time (h)
$\bar{v}_n, \underline{v}_n$ maximum and minimum technical water storage limits of node n which is a reservoir $e(n)$ due to design characteristics (h m^3)
$\bar{v}_{n,p}$, $\underline{v}_{n,p}$ maximum and minimum water storage limit imposed by water authorities in node n which is a reservoir $e(n)$ and period p (h m^3)
vf_n target water volume of node n which is a reservoir $e(n)$ (h m^3)
$VUP_{w,n,p}, VDW_{w,n,p}$ maximum and minimum water volume in node $n \in E(n)$ to calculate hydropower generation curves (h m^3)
$VM_{w,n,p}$ mean value of $VUP_{w,n,p}$ and $VDW_{w,n,p}$ (h m^3)
ρ_w probability of scenario w

Positive variables

$aux_b_{w,g,p}$ auxiliary variable to obtain a linear expresion of the head-dependent secondary regulation-down reserve (MW)
$b^{up}_{w,g,p}, b^{dw}_{w,g,p}$ secondary regulation-up and down reserve in hydropower unit g (MW)
$hg_{w,g,p}$ hydropower generation in unit g (MW)
$hg^{+}_{w,g,p}$, $hg^{-}_{w,g,p}$ power generation and consumption in pumped storage hydropower unit $g \in$ PG (MW)
$q_{w,g,p}$ water discharged flow in hydropower unit g (h m^3/h)
$q^{+}_{w,g,p}, q^{-}_{w,g,p}$ water discharged and pumped water flow in pumped storage hydropower unit $g \in PG$ (h m^3/h)
$q_{w,j,p}$ water discharged flow through arc j (h m^3/h)
$qs_{w,g,p}$ water discharged flow in a hydropower unit g above its minimum technical limit (h m^3/h)
$s_{w,j,p}$ water spilled flow through arc j (h m^3/h)
$v_{w,n,p}$ water storage volume (h m^3)
$y_{w,g,p}$ start-up decision (start-up=1/other=0)
$z_{w,g,p}$ shut-down decision (shut-down=1/other=0)

Binary variables

$d_{w,e,p}$ auxiliary binary variable used for the discretization of the hydropower generation curve in reservoir e
$ub_{w,g,p}$ state of hydropower unit g for pumping mode (On=1/Off=0)
$ut_{w,g,p}$ state of hydropower unit g for generating mode (On=1/Off=0)

guaranteed to be available in case they are requested in real-time by the Transmission System Operator (TSO). Other articles, such as [4], model how to obtain the optimal hydropower scheduling participating in day-ahead electricity market and ancillary services markets but only for an isolated pumped storage plant, without taking into account effects in a multi-reservoir cascaded system. Similarly to [3], the stochasticity about market prices is considered in [5] although the future uncertainty about the water inflows is not introduced. Analogously to [4], the formulation presented in [5] does not take into account complex real-life hydro system topologies, so its optimal solution cannot be used in the operation of real-life hydro systems, with several river reaches feeding the same reservoir or several canals using water from a certain reservoir. In addition to this, to authors' knowledge there are no published models that formulate an explicit expression to compute secondary regulation-down reserve in a head-dependency power generation framework.

This paper presents a stochastic optimization model which maximizes the expected profit of a real-life hydropower generation company which participates in the day-ahead electricity market and the secondary regulation reserve market (secondary regulation refers to the participation in the automatic generation control, AGC) of the Spanish power system. The time horizon of the model is one week and it is based on Mixed Integer Linear Programming (MILP). The formulation takes into account the existence of several operational constraints imposed by the corresponding water authority, such as water outflows for irrigation, stored water limits for purposes different from hydroelectricity production (e.g. navigation) or water released limits in river reaches in order to protect downstream flora and fauna.

The proposed model also considers the existence of long-term guidelines or water volume signals in the form of end of week target stored volume in reservoirs [6], and the loss of efficiency derived from the need of operating the unit out of the maximum efficiency range in order to be able to provide regulation reserves. As a consequence of this lower efficiency, the spot market revenues will be also affected, and this effect is also taken into account similarly as in [7]. In addition to this, uncertainty has been taken into account by means of a stochastic tree where considered random parameters are water inflows and market prices.

Main contributions of this paper with respect to previous works can be summarized as follows. Firstly, as far as authors know, there is no other hydro scheduling model where the following aspects have been considered altogether: precise and very detailed representation of the hydro system topology, existence of pumped storage power plants, fractional water travel time, limits on maximum start-up and shut-down maneuvers per day, minimum on/off time at unit level, head dependent unit generation characteristics, day-ahead electricity market and secondary regulation reserve market participation and, finally, water inflows and market prices uncertainty. The inherent difficulties to consider all these elements jointly have been overcome by an adequate formulation, as it will be shown in the realistic case study. Secondly, a MILP formulation to model secondary regulation-up and down reserves, taking into account hydropower generation head-dependency is presented. Finally, the main contribution of this paper is that the obtained solution protects a hydropower producer against risk of water unavailability due to the uncertainty in the real-time use of both the secondary regulation-up and down reserves.

As it will be shown later, this is achieved by a set of linear constraints included in the MILP formulation. As the market of operating reserves is becoming increasingly important, the proposed new formulation can help the hydro producers to assess more accurately the amount of operating reserves that can be actually offered in the market. This can reduce the risk of hydro producers of being penalized when the cleared regulation reserves cannot be deployed in real time due to having reached the storage water limits as it will be explained in Section 2.2.5.

The paper has been structured as follows: Section 2 refers to model formulation, composed of objective function and the constraints. The case study is analyzed in Section 3. Then, conclusions are presented in Section 4. Finally, future work appear in Section 5.

2. Model formulation

2.1. Objective Function

The main goal of the presented model is to maximize the expected revenues of a price-taker hydropower company for participating simultaneously in the day-ahead electricity market and the secondary regulation reserve market (1).

$$\text{Max} \quad \sum_{w} \rho_w \cdot \left(\text{rev}_w^{\text{EM}} + \text{rev}_w^{\text{SM}}\right) \tag{1}$$

Expected revenues in the day-ahead electricity market and secondary regulation reserve market are defined in (2) and (3), respectively.

$$\text{rev}_w^{\text{EM}} = \sum_{p} l_p \cdot \lambda\text{EM}_{w,p} \cdot \sum_{g} hg_{w,g,p} \tag{2}$$

$$\text{rev}_w^{\text{SM}} = \sum_{p} \lambda\text{SM}_{w,p} \cdot \sum_{g \in \text{BG}} \left(b_{w,g,p}^{up} + b_{w,g,p}^{dw}\right) \tag{3}$$

2.2. The model constraints

2.2.1. The water balance equation

A real-life hydro system is formed by several items such as reservoirs, hydropower plants, water inflows, canals and river reaches. In order to model in detailed a hydro system, different topology representations have been described in the literature. The presented model adapts the features proposed by [8], in which two main modeling entities have been used: *nodes* and *arcs*. In our case, the former correspond to reservoirs, hydropower plants and points of the system where there are water inflows/outflows. The latter are used to connect nodes (for example, canals or river reaches) and are classified into two types: discharge water arcs and spillage arcs. All sorts of nodes and arcs with most of their parameters and variables are presented graphically in Fig. 1. The proposed hydro system representation is mathematically compact, easy to implement and flexible enough to model in detail multi-reservoir systems.

The uncertainty about water inflows and market prices is modeled in this paper through a finite number of scenarios, forming a stochastic scenario tree. The scenario tree is composed by a single scenario during the first day of the week and several scenarios for the rest of the days (Fig. 2 shows an example of the structure of the proposed scenario tree with a single scenario during the first three periods and three scenarios for the rest of the periods). The authors consider such a scenario tree structure in order to obtain a single decision the first day taking into account the unknown future of the rest of the week (bidding strategies are outside the scope of this article). The scenario tree is split according to *Type* 1 in [9], based on father and son scenarios. For example, w_1 scenario is established as the father of w_2 and w_3 scenarios in the scenario tree of Fig. 2. The scenario tree structure has been decided in order to be consistent with the daily execution of the model. Decisions corresponding to the first 24 h can be considered as here-and-now decisions, whereas the following 144 h can be considered as a "look ahead" period, [10] which may be updated as new prices and water inflows forecasts become available.

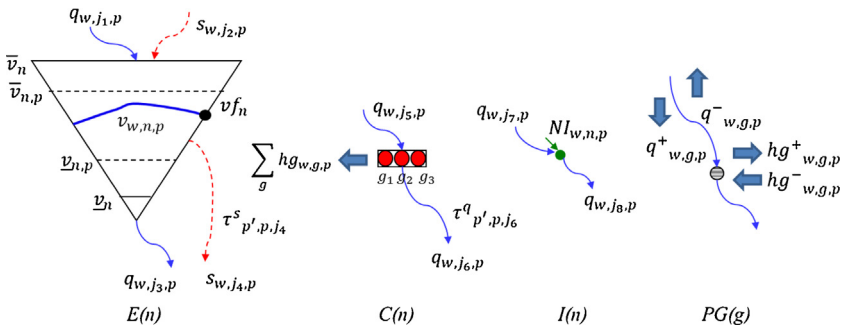


Fig. 1. Nomenclature.

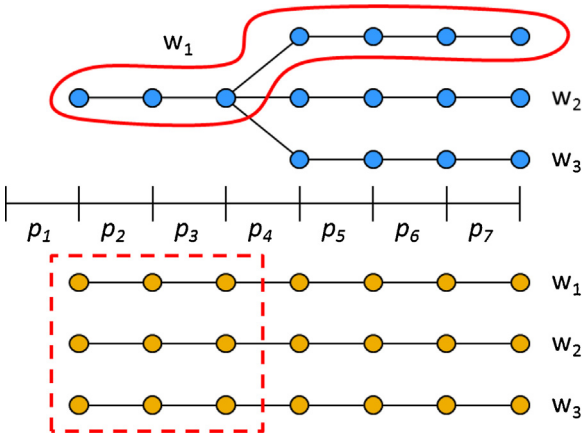


Fig. 2. Treatment of stochastic scenario tree.

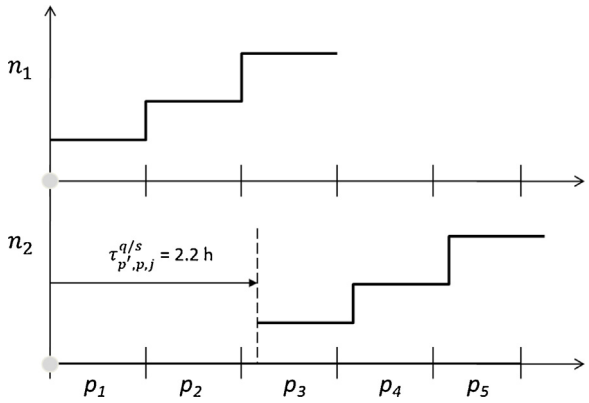


Fig. 3. Explanation of water travel time modeling. Water discharged from node n_1 in period p. Water arriving at node n_2 from node n_1 in period p.

The balance equation, which takes into account the scenario tree of future uncertainty and the fractional water travel time between nodes is presented in (4).

$$\frac{1}{l_p}\cdot\left(v_{w,e(n),p}-v_{w'(p-1),e(n),p-1}\right) = NI_{w,n,p}+\sum_{p'\le p}\left(\sum_{j\in Uq(n)} q_{w'(p'),j,p'}\cdot\tau^{q}_{p',p,j}+\sum_{j\in Us(n)} s_{w'(p'),j,p'}\cdot\tau^{s}_{p',p,j}\right) - \left(\sum_{j\in Dq(n)} q_{w,j,p}+\sum_{j\in Ds(n)} s_{w,j,p}\right) \qquad \forall(w,n,p) \tag{4}$$

The interpretation of (4) is the following. In a certain scenario W and period P, the difference between the water volume in a node N which is a reservoir e minus its volume in the previous period equals to the sum of the water inflows/outflows to/from node N, the discharged water flow from upper nodes $Uq(n)$ in periods P', affected by the parameter $\tau^{q}_{p',p,j}$, the spilled water flow from upper nodes $Us(n)$ in periods P', affected by the parameter $\tau^{s}_{p',p,j}$, minus the discharged and spilled water flow from node N. The balance equation introduces temporal links between branches of the scenario tree because volume and water flows in previous periods could belong to a different branch. The presented formulation selects the scenario for previous periods, W', as the result of dependency of previous period, $p-1$ or P' and current scenario, W.

Fractional water travel time is modeled using parameters $\tau^{q}_{p',p,j}$ and $\tau^{s}_{p',p,j}$. They measure the percentage of discharged and spilled water flow, respectively, through arc J, from an upper node in period P' reaching the lower node in period P, depending on its water travel time. For instance, a fractional water travel time of 2.2 h is shown in Fig. 3: 20% of flow released from node n_1 in period p_2 and 80% in p_3 arrive in node n_2 in period p_5 ($\tau^{q}_{p_2,p_5,j}=0.2$ and $\tau^{q}_{p_3,p_5,j}=0.8$). $\tau^{s}_{p',p,j}$ is computed analogously. The topology of the hydro system under study obliges to distinguish between $\tau^{q}_{p',p,j}$ and $\tau^{s}_{p',p,j}$ since discharge and spillage arcs connected between the same two nodes could have different water travel times, as it occurs in a large number of existing hydro systems.

2.2.2. Hydropower unit generation

Several techniques have been proposed in the literature in order to model head-dependency in hydropower production curves. In [11], a triangularization of the input–output surface is proposed, applying MILP. Due to the large quantity of binary variables required, computational burden can be intractable in systems with multi-hydropower plants. On the other hand, the iterative algorithm proposed in [12] does not guarantee a global optimal solution of the problem, [13]. Other interesting approaches based on non-linear programming (NLP), [14] and on mixed integer non-linear programming (MINLP), [15] have not been assessed since the consideration of several scenarios may result in unacceptable CPU times. In addition, the use of MINLP commercial solvers, such as the one used in [15] for solving such a complex problem must be done with the due caution, [16]. Tailored MINLP solution algorithms, such as the one proposed in [16], were not tested due to both their CPU times and complexity.

Consequently, the presented formulation follows the one proposed by [17]. In order to reduce the computational burden, two power-discharge curves (A and B) in each scenario, period and hydropower unit are used in the model, Fig. 4. Power-discharge

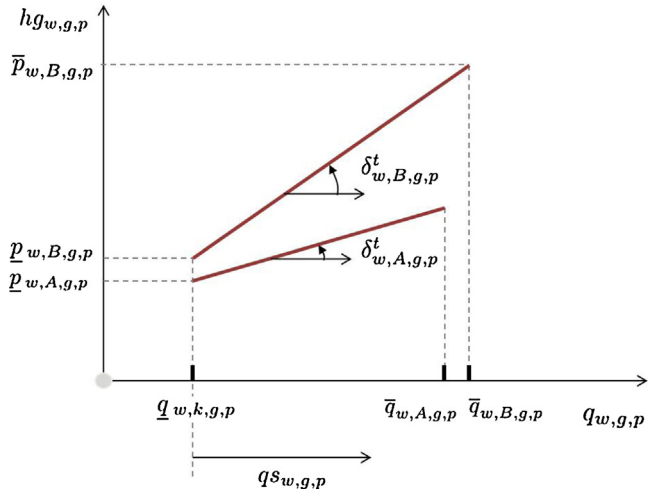


Fig. 4. Hydropower unit generation.

curves and generating energy coefficients of each unit, $\delta^t_{w,A,g,p}$ and $\delta^t_{w,B,g,p}$, which also change in each scenario and period, depending on natural water inflows, are calculated in a previous process. This process obtains maximum, $VUP_{w,n,p}$ and minimum, $VDW_{w,n,p}$ theoretical water volumes in the following manner: maximum volume in period p equals the sum of maximum volume in previous period, stochastic inflow and maximum discharged volume from upper reservoirs, and minimum volume in period p equals the sum of minimum volume in previous period and stochastic inflow minus maximum discharged flow to lower reservoirs. Both maximum and minimum theoretical volumes are limited by maximum and minimum technical storage, respectively. The net head related to Curve A (resp. B) is the corresponding to a water volume of the mean value between $\mathrm{VM}_{w,n,p}$ and $VDW_{w,n,p}$ (resp. $VUP_{w,n,p}$ and $\mathrm{VM}_{w,n,p}$). All technical data can be obtained directly from several power plant data [13].

From the modeling point of view, power generation (7)–(10) and head-dependent maximum water flow (12) are selected depending on the value of the binary variable $d_{w,e,p}$, which depends on the water volume of associated reservoir, (5) and (6). In addition to this, Eqs. (11), (13) and (14) ensure that when a hydropower unit is off, there is no energy generation nor releasing water flow (note that minimum water flow is the same in both curves, A and B, $\underline{q}_{w,A,g,p} = \underline{q}_{w,B,g,p}$).

A hydropower unit usually requires to be a minimum on/off time when it is started-up/shut-down [8]. Both cases have been modeled according to the formulation used in [18] for thermal units (15) and (16). Start-up and shutdown decisions are coupled to the unit's state variable in a period P and previous period, $p-1$ in (17). Besides, the model protects the turbines' lifespan or deems crew constraints by limiting the maximum daily start-up and shutdown maneuvers per day according to OP_g (18). Finally, water discharged through all units belonging to the same hydropower plant equals the discharged water flow of the latter through the corresponding arc (19).

$$v_{w,n,p} \le \mathrm{VUP}_{w,n,p} \cdot d_{w,e(n),p} + \mathrm{VM}_{w,n,p} \cdot \left(1 - d_{w,e(n),p}\right) \quad \forall (w, n \in E, p) \tag{5}$$

$$v_{w,n,p} \ge \mathrm{VDW}_{w,n,p} \cdot \left(1 - d_{w,e(n),p}\right) + \mathrm{VM}_{w,n,p} \cdot d_{w,e(n),p} \quad \forall (w, n \in E, p) \tag{6}$$

$$hg_{w,g,p} \le \underline{p}_{w,A,g,p} \cdot ut_{w,g,p} + qs_{w,g,p} \cdot \delta^t_{w,A,g,p} + M \cdot d_{w,eg(g),p} \quad \forall (w, g, p) \tag{7}$$

$$hg_{w,g,p} \ge \underline{p}_{w,A,g,p} \cdot ut_{w,g,p} + qs_{w,g,p} \cdot \delta^t_{w,A,g,p} - M \cdot d_{w,eg(g),p} \quad \forall (w, g, p) \tag{8}$$

$$hg_{w,g,p} \le \underline{p}_{w,B,g,p} \cdot ut_{w,g,p} + qs_{w,g,p} \cdot \delta^t_{w,B,g,p} + M \cdot \left(1 - d_{w,eg(g),p}\right) \quad \forall (w, g, p) \tag{9}$$

$$hg_{w,g,p} \ge \underline{p}_{w,B,g,p} \cdot ut_{w,g,p} + qs_{w,g,p} \cdot \delta^t_{w,B,g,p} - M \cdot \left(1 - d_{w,eg(g),p}\right) \quad \forall (w, g, p) \tag{10}$$

$$q_{w,g,p} = \underline{q}_{w,A,g,p} \cdot ut_{w,g,p} + qs_{w,g,p} \quad \forall (w, g, p) \tag{11}$$

$$qs_{w,g,p} \le \left(\bar{q}_{w,A,g,p} - \underline{q}_{w,A,g,p}\right) \cdot \left(1 - d_{w,eg(g),p}\right) + \left(\bar{q}_{w,B,g,p} - \underline{q}_{w,B,g,p}\right) \cdot d_{w,eg(g),p} \quad \forall (w, g, p) \tag{12}$$

$$qs_{w,g,p} \le ut_{w,g,p} \cdot \bar{q}_g \quad \forall (w, g, p) \tag{13}$$

$$hg_{w,g,p} \le ut_{w,g,p} \cdot \bar{p}_{w,B,g,p} \quad \forall (w, g, p) \tag{14}$$

$$\sum_{p'=p-TU_g+1}^{p} y_{w,g,p'} \le ut_{w,g,p} \quad \forall \left(w, g, p \in \left[TU_g, \mathrm{P}\right]\right) \tag{15}$$

$$\sum_{p'=p-TD_g+1}^{p} z_{w,g,p'} \le 1 - ut_{w,g,p} \quad \forall \left(w, g, p \in \left[TD_g, \mathrm{P}\right]\right) \tag{16}$$

$$ut_{w,g,p} = ut_{w,g,p-1} + y_{w,g,p} - z_{w,g,p} \quad \forall (w, g, p) \tag{17}$$

$$\sum_{p \in day} \left(y_{w,g,p} + z_{w,g,p}\right) \le OP_g \quad \forall (w, g, day) \tag{18}$$

$$\sum_{j \in Dq(n)} q_{w,j,p} = \sum_{g \in CG(n)} q_{w,g,p} \quad \forall (w, n \in C, p) \tag{19}$$

2.2.3. Pumped storage units

A pumped storage unit, *pg*, can be utilized for generating and consuming energy. The former operation (referred to variables $q^+_{w,pg,p}$ and $hg^+_{w,pg,p}$ in the presented formulation in (20) and (21)) has been implemented analogously to Section 2.2.2.

The water flow and power generation of a *pg* unit have been split in two parts: the discharged water flow (generation mode) and the pumped water flow (consumption mode), (20), (21) and Fig. 5. Proposed *pg* units must be operated at a fixed point in pumping mode. Therefore, when it is on ($ub_{w,g,p} = 1$), pumped water flow is fixed at the maximum value, $\bar{q}^b_g$ (22), consuming the maximum power in pumping mode, $\bar{p}^b_g$ (23). In order to be coherent with variable definitions (positive variables), lower bounds of water discharged flow, $q_{w,g,p}$ and power generation, $hg_{w,g,p}$ in *pg* units are established to minus their maximum pumped water flow and minus their maximum consumed power, respectively. Finally, the presented model does not permit any *pg* unit to discharge and pump water flow simultaneously (24).

$$q_{w,g,p} = q^+_{w,g,p} - q^-_{w,g,p} \quad \forall (w, g \in \mathrm{PG}, p) \tag{20}$$

$$hg_{w,g,p} = hg^+_{w,g,p} - hg^-_{w,g,p} \quad \forall (w, g \in \mathrm{PG}, p) \tag{21}$$

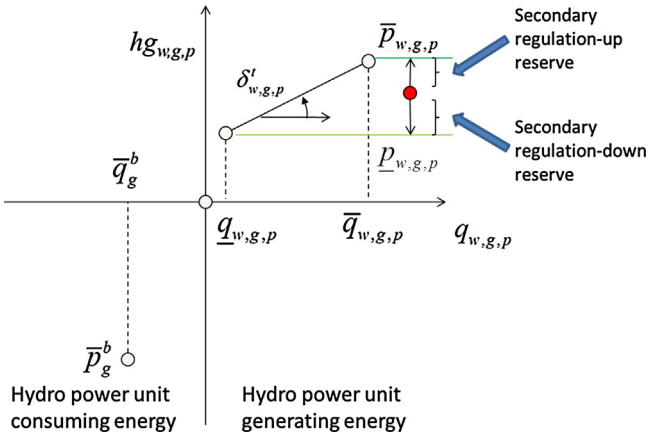


Fig. 5. Generation in pumped storage hydropower units.

$$q^{-}_{w,g,p} = ub_{w,g,p} \cdot \bar{q}^b_g \qquad \forall (w, g \in \mathrm{PG}, p) \tag{22}$$

$$hg^{-}{}_{w,g,p} = ub_{w,g,p} \cdot \bar{p}^b_g \qquad \forall (w, g \in \mathrm{PG}, p) \tag{23}$$

$$ut_{w,g,p} + ub_{w,g,p} \leq 1 \qquad \forall (w, g \in \mathrm{PG}, p) \tag{24}$$

2.2.4. General hydraulic constraints

The hydro system management is subject to operational constraints, affecting the optimal solution. Typically, these constraints are imposed by water authorities in order to protect the environment and hydraulic usage (25) and (26) or set in order to take into account the results of a longer term scheduling procedure, through target water volume vf_n at the last period of the time horizon, P (27).

$$\max\left(\underline{v}_{n,p}, \underline{v}_n\right) \leq v_{w,n,p} \leq \min\left(\bar{v}_{n,p}, \bar{v}_n\right) \qquad \forall (w, n \in E, p) \tag{25}$$

$$\underline{rf}_{n,p} \leq \left(\sum_{j \in Dq(n)} q_{w,j,p} + \sum_{j \in Ds(n)} s_{w,j,p}\right) \leq \bar{rf}_{n,p} \qquad \forall (w, n \in (E \cup C), p) \tag{26}$$

$$v_{w,n,\mathrm{P}} = vf_n \qquad \forall (w, n \in E) \tag{27}$$

2.2.5. Secondary regulation reserve in hydropower units

When a *bg* unit is generating energy, i.e. $ut_{w,g,p} = 1$, a hydropower decision maker must decide what quantity of its available regulation-up and down reserve capacity is offered in the secondary reserve market. The offered secondary regulation-up reserve must be lower or equal to the difference between the head-dependent maximum power production, $\bar{p}_{w,k,g,p}$ and its actual generation level (28). Similarly, its secondary regulation-down reserve must be lower or equal to the difference between its generation level and the head-dependent minimum power production, $\underline{p}_{w,k,g,p}$, (29). If a unit *bg* is shut down, its available secondary regulation reserve must be zero, (30).

The auxiliary variable $aux_b_{w,g,p}$ is proposed in the presented model to avoid non-linearities in secondary regulation reserve formulation due to head-dependency in hydropower production. When a *bg* unit is off, $aux_b_{w,g,p}$ variable equals the expression in square brackets in (29) and when the unit is on, $aux_b_{w,g,p}$ variable equals zero through (31).

In the presented model, we assume that pumped storage hydrounits do not participate in the secondary regulation reserve market. As it is pointed out in [1], this is a promising area of future research.

Table 1
Nominal data of reservoirs and hydropower plants.

Node Unit	$\bar{v}_n$ h m³	$\underline{v}_n$ h m³	$\bar{q}_n$ m³/s	$\underline{q}_n$ m³/s	$\bar{p}_n$ MW	$\underline{p}_n$ MW
E_1	75	5	250	0	–	–
E_2	15	3	250	−200	–	–
E_3	1000	30	1000	0	–	–
C_1	–	–	250	90	150	48.6
C_2	–	–	250	−200	200	−250
C_3	–	–	1 000	120	500	54

$$b^{up}_{w,g,p} \leq \left[d_{w,eg(g),p} \cdot \bar{p}_{w,B,g,p} + \left(1 - d_{w,eg(g),p}\right) \cdot \bar{p}_{w,A,g,p}\right] - hg_{w,g,p} \qquad \forall (w, g \in \mathrm{BG}, p) \tag{28}$$

$$b^{dw}_{w,g,p} - aux_b_{w,g,p} \leq hg_{w,g,p} - \left[d_{w,eg(g),p} \cdot \underline{p}_{w,B,g,p} + \left(1 - d_{w,eg(g),p}\right) \cdot \underline{p}_{w,A,g,p}\right] \qquad \forall (w, g \in \mathrm{BG}, p) \tag{29}$$

$$b^{up}_{w,g,p} + b^{dw}_{w,g,p} \leq M \cdot ut_{w,g,p} \qquad \forall (w, g \in \mathrm{BG}, p) \tag{30}$$

$$aux_b_{w,g,p} \leq M \cdot \left(1 - ut_{w,g,p}\right) \qquad \forall (w, g \in \mathrm{BG}, p) \tag{31}$$

Hydro power is considered an energy-constrained generation technology because the amount of stored water in the reservoirs is limited. These limits can be physical (for instance, due to the maximum reservoir storage capacity and the quantity of natural inflows received during a certain period of time), or they can be the consequence of a given policy (for instance, the reservoir level limits set by the water authorities in charge of the flood/drought control and management). In hydro systems formed by multiple small-sized reservoirs, the usage of the scheduled reserves could represent a substantial percentage of the stored energy. Therefore, it would be necessary to protect a hydropower producer against risk of (1) water unavailability due to the uncertainty in the real-time use of the secondary regulation-up reserves and (2) storage unavailability due to the uncertainty in the real-time use of the regulation-down reserves. This issue has not been treated previously in the related literature and it constitutes one of the contributions of this work. In order to assess the effect of requesting the reserves, it is necessary to estimate how much of the offered reserves are going to be requested by the TSO in real-time to carry out the active power balance and load following. As the use of the available reserves is subject to uncertainty, in this paper it is proposed to introduce the estimated percentage of use of the available reserves as an input parameter. These percentages can be estimated by statistical analysis of historical data, although this estimation is out of the scope of this paper.

In order to obtain a conservative solution, it will be considered that in every period P, the $\mathcal{F}^{\mathrm{up}}$ percentage of the scheduled regulation-up reserve, and the $\mathcal{F}^{dw}$ percentage of the scheduled regulation-down reserves will be used by the TSO. The consideration of both "contingencies" in the planning stage, makes the obtained schedule robust against both situations that will seldom take place simultaneously. The mathematical formulation proposed in this paper to protect a hydropower producer against risk of water unavailability due to uncertain real-time use of reserves, is the one shown in equations (32) and (33). Both equations do not usually appear in the framework of thermal power plants because

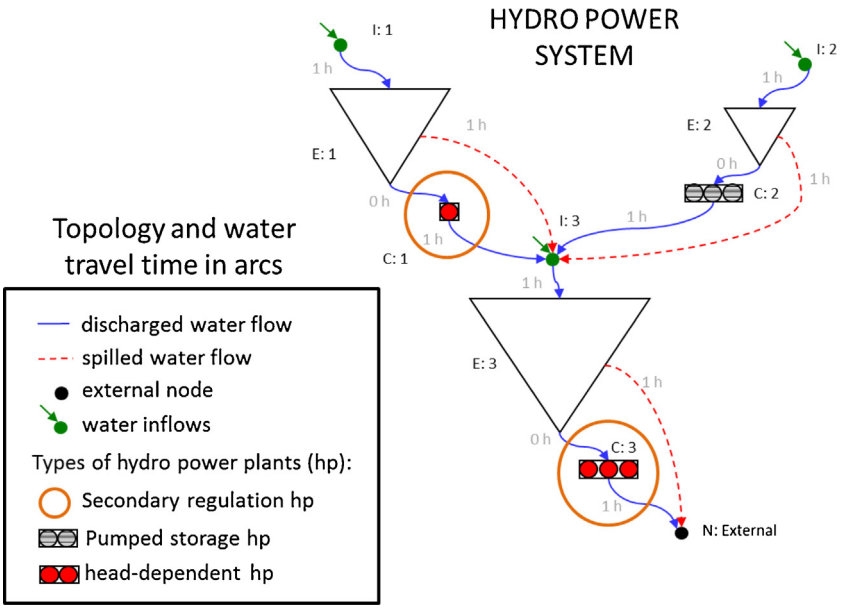


Fig. 6. Hydro system proposed in the case study.

their fuel is considered unlimited in the short-term in contrast to hydropower technology.

$$\sum_{n' \in UPC(n)} \left(v_{w,e(n'),p} - \max\left(\underline{v}_{n'}, \underline{v}_{n',p} \right) \right) \geq \frac{\mathcal{F}^{up}}{100} \cdot \sum_{p' \leq p} \sum_{g \in (CG(n) \cap BG(g))} \left(l_p \cdot \frac{b^{up}_{w'(p'),g,p'}}{\min_k \left(\delta^t_{w'(p'),k,g,p'} \right)} \right) \quad \forall (w, n \in C, p) \tag{32}$$

Notice that Eq. (32) is defined for every scenario W, every hydropower plant N, and every time period P. Its left hand side represents the amount of water that might have been released up to that period without violating the most restrictive minimum-volume constraint. The right hand side computes the amount of water that would be released from the first hour of the scheduling until the period for which the constraint is formulated in case the TSO makes use of a given percentage $\mathcal{F}^{up}$ of the scheduled regulation-up reserve. As the regulation-up reserve is expressed in terms of power $b^{up}_{w,g,p}$, it is transformed into water volume by multiplying it by the duration of the period l_p, and by dividing it by the generating energy coefficient of that unit $\delta^t_{w,k,g,p}$. Notice that the minimum value of $\delta^t_{w,k,g,p}$ is chosen as this is the one that results in the higher water consumption value, leading again to a conservative solution.

$$\sum_{n' \in UPC(n)} \left[\min\left(\bar{v}_{n'}, \bar{v}_{n',p} \right) - v_{w,e(n'),p} \right] \geq \frac{\mathcal{F}^{dw}}{100} \cdot \sum_{p' \leq p} \sum_{g \in (CG(n) \cap BG(g))} \left(l_p \cdot \frac{b^{dw}_{w'(p'),g,p'}}{\min_k \left(\delta^t_{w'(p'),k,g,p'} \right)} \right) \quad \forall (w, n \in C, p) \tag{33}$$

An analogous reasoning can be applied in Eq. (33) where its left hand side represents the amount of water that might have been kept up to that period without violating the most restrictive maximum-volume constraint. Therefore, in this case the solution protects a hydropower producer against risk of storage unavailability due to uncertain real-time use of regulation-down reserves, when the hydropower plant is requested to decrease its output power.

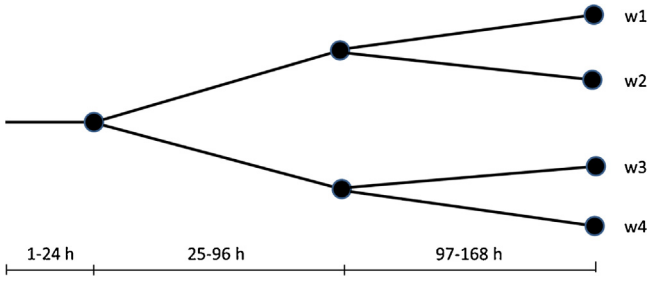


Fig. 7. Scenario tree to model uncertainty in water inflows and market prices.

3. Case study

The hourly scheduling of the hydro system shown in Fig. 6 has been optimized by the proposed model on a weekly time horizon. The time horizon covers seven days from 15th to 21th of March in 2013. The system is composed by nine nodes: three reservoirs, three natural water inflow points and three hydropower plants. Two hydropower plants have *bg* units and there is one pumped storage plant. The nominal data of each reservoir and hydropower plant are presented in Table 1.

The uncertainty in the case study has been considered in water inflows, the day-ahead electricity market prices and the secondary regulation reserve market prices. This uncertainty has been modeled through a scenario tree composed by four branches, Fig. 7. The first 24 h have the same information so all the branches share a common scenario. In the following three days, the scenario tree is branched in two scenarios. Finally, in last three days, each branch is divided into other two scenarios. All scenarios have been assigned the same probability.

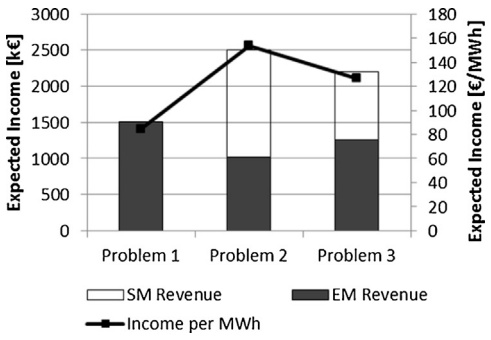


Fig. 8. Expected Income and expected income per MWh for participating in the proposed electricity markets.

Three different strategies are considered:

- Strategy 1: optimal hydro scheduling for participating only in the day-ahead electricity market. The available secondary regulation-up and down reserves computed as an ex-post result (not optimized), are also shown to assess the expected incomes in the secondary regulation market.
- Strategy 2: optimal hydro scheduling for participating both in the day-ahead electricity market and in the secondary reserve market without taking into account the limits imposed by (32) and (33). Notice that Strategy 2 corresponds to consider $\mathcal{F}^{up}$ and $\mathcal{F}^{dw}$ parameters equal 0 % in (32) and (33).
  - Strategy 2bis: Same as 2, but without considering the net head effect on the values of the regulation-up and down reserves, and neither the water travel times.
- Strategy 3: optimal hydro scheduling for participating both in the day-ahead electricity market and in the secondary reserve market taking into account the complete model proposed in this paper. Note that with this strategy, a decision maker prefers to be more conservative than in Strategy 2, using $\mathcal{F}^{up}$ and $\mathcal{F}^{dw}$ parameters equal 100% in (32) and (33).

All MILP strategies have been optimized in a computer with 3.1 GHz Intel(R) Xeon(R) CPU, with 64 GB of RAM memory, with GAMS software and Cplex 12.2 solver, which uses a branch and cut algorithm to solve MILP problems.

As shown in Fig. 8 and Table 2, expected total income and expected total income per MWh (the latter is the mean value of total income divided by EM energy in each scenario) in Strategy 2 are higher than in Strategy 1. Although the optimal solution in Strategy 1 does not take into account the reserve market, all available reserves according to the operation schedule obtained in Strategy 1, are valued at the reserve market prices in an ex-post analysis. Strategy 1 would obtain an extra expected income of 256,957 € if the hydropower producer offers all available reserves. And, therefore, total expected income in both markets will be 1,764,203 € with an expected income per MWh of 97.63 €/MWh. In Strategy 2, the expected income per MWh is around 55% higher than in Strategy 1 in this case study, which represents a significant increase in expected income per MWh and highlights the importance of considering the reserve market within the model. But also, expected total income is 13% higher and expected total income per MWh is around 21% higher in Strategy 2 than in Strategy 3. However, as it has mentioned above, the latter introduces constraints (32) and (33) with $\mathcal{F}^{up}$=$\mathcal{F}^{dw}$= 100%. This means the model assumes that 100% of secondary regulation-up and down reserves will be used simultaneously in all hours. Therefore, availability in reserves is reduced and, hence, expected income from the reserve market is also reduced. But at the same time, Strategy 3 offers more energy in the spot market and obtains more expected income in this market, compensating loss of expected profit in the reserve market comparing to Strategy 2. In addition to this, Strategy 3 minimises the risk of deviations of providing secondary regulation-up and down energy requested by the TSO in real-time. Nevertheless, participating in both markets with the solution of Strategy 2 is highly risky as all available secondary regulation reserves are offered without taking into account water and storage availability. For that reason, although the expected total income and expected income per MWh in Strategy 3 are lower that in Strategy 2, that solution protects more a hydropower producer against risk of water and storage unavailability due to the uncertainty in the real-time use of both the secondary regulation-up and down reserves. Notice that in the Spanish electricity market although secondary regulation reserves must fulfil a rate between regulation-up and down reserves, this constraint is included neither in the model nor in the results of the case study in order to present a more general case. The optimization problem corresponding to Strategy 3 is composed by 56,255 constraints and 39,673 variables (5928 of them are binary variables), and required a computational time consumption of around 10 min.

The solution provided by the model is also composed by the water volume of each reservoir, the hydropower schedule of each unit and its secondary regulation-up and down reserve schedule and, finally, the spilled water flow in each node.

The aggregated power generation of the hydro system and secondary regulation reserves obtained in w_1 scenario of Strategies 1, 2 and 3 are presented in Figs. 9–11,,, respectively. Fig. 9 also shows spot prices in w_1 scenario and Figs. 10 and 11 show reserve prices in w_1 scenario. As available secondary regulation reserves are more limited in Strategy 3 than Strategy 2, the criterion of expected profit maximization changes the offered reserves between Strategy

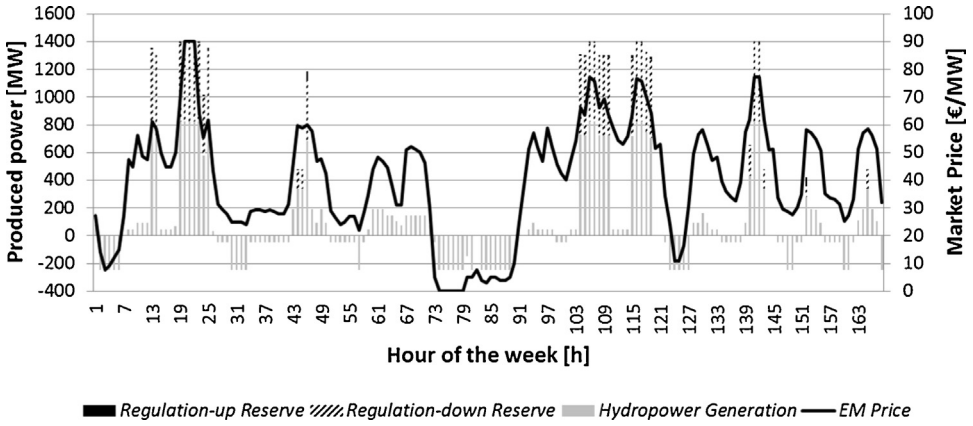


Fig. 9. Hydro scheduling and spot price with Strategy 1 in scenario w_1.

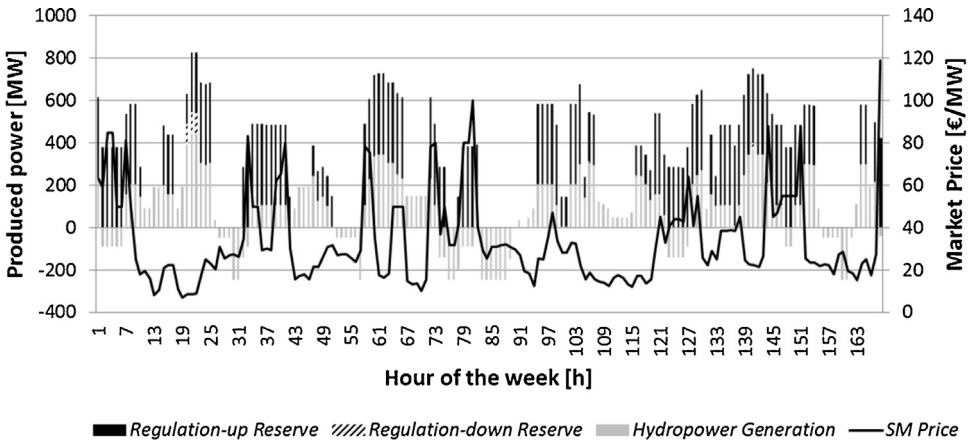


Fig. 10. Hydro scheduling and secondary reserve and price with Strategy 2 in scenario w_1.

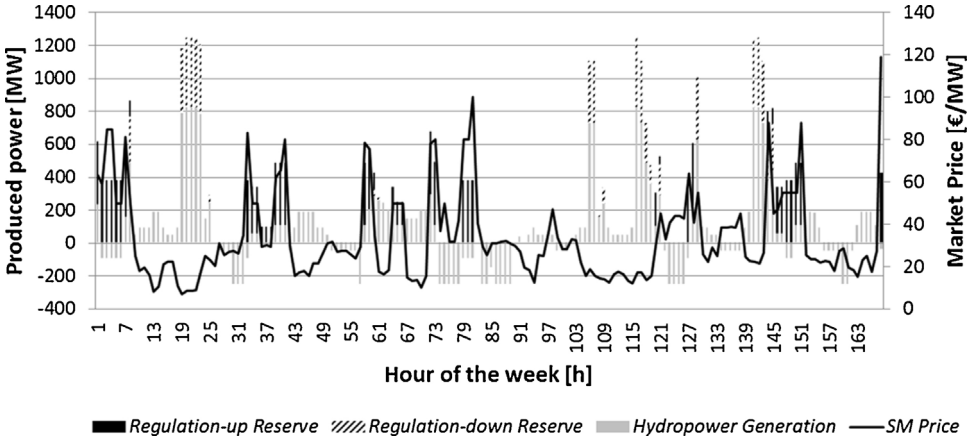


Fig. 11. Hydro scheduling and secondary reserve and price with Strategy 3 in scenario w_1.

2 and 3. For instance, the offered reserves during periods 31–40 and 142–151 are more adjusted to the reserve price profile in Strategy 3 than in Strategy 2.

In Fig. 12, both the water flow of the pumped storage power plant (C_2) and the evolution of the stored volume of the upper reservoir (E_2) in w_1 scenario of strategy 3 are shown. Since reservoir E_2 is located at the head of the hydro chain, the evolution of the stored volume is determined by the operation of the plant and the natural water inflows. In Fig. 13, the generation and consumption schedule of each pumped storage hydropower unit in w_1 scenario of Strategy 3 and the day-ahead electricity market price in the same scenario are presented. The obtained power schedule follows a rational operation of income maximization, discharging water flow and generating energy when the day-ahead market prices are high and pumping water flow and consuming energy when prices are lower. In the latter periods, pumped storage units of C_2 consume their fixed maximum power, $\bar{p}_g^b$, in this case, 100 *MW* in G_1 and G_2 and 50 *MW* in G_3.

Solution in Strategy 2 is compared to the solution obtained in Strategy 2bis, which does not consider fractional water travel times and head-dependency in power generation and reserves. According to Table 2, Strategy 2bis obtains 58274 € less than Strategy 2 (expected income of Strategy 2 is 2.39% higher than in Strategy 2bis) as it produces 931 MWh less in the spot market and it offers 1677.91 MW less in the reserve market than Strategy 2. The reason for this is head-dependency because power generation curves in Strategy 2bis does not change irrespective of the head in reservoirs (a power generation curve related to the head of the medium water volume is considered) whereas reservoirs in Strategy 2 are operated taking into account that a higher head means higher production and reserves, at the same time (more reserves because maximum power generation is larger as head increases). In addition to this, solution in Strategy 2 is not only better than in Strategy 2bis from an economic point of view but also because it is more realistic than in Strategy 2bis as fractional water travel times are deemed.

Table 2
Expected results of Strategies 1–3.

	Strategy 1	Strategy 2	Strategy 2bis	Strategy 3
EM Energy (MWh)	18 070.59	16 497.25	15 566.25	17 626.99
SM Reserve (MW)	16055	38 617.19	36 939.28	19 971.74
EM Revenue (€)	1 507 246	1 021 568	999 206.56	1 254 552
SM Revenue (€)	256 957	1 477 216	1 441 304	947 760
Total revenue (€)	1 764 203	2 498 784	2 440 510	2 202 312
Income (€/MWh)	98.87	153.68	159.56	126.63

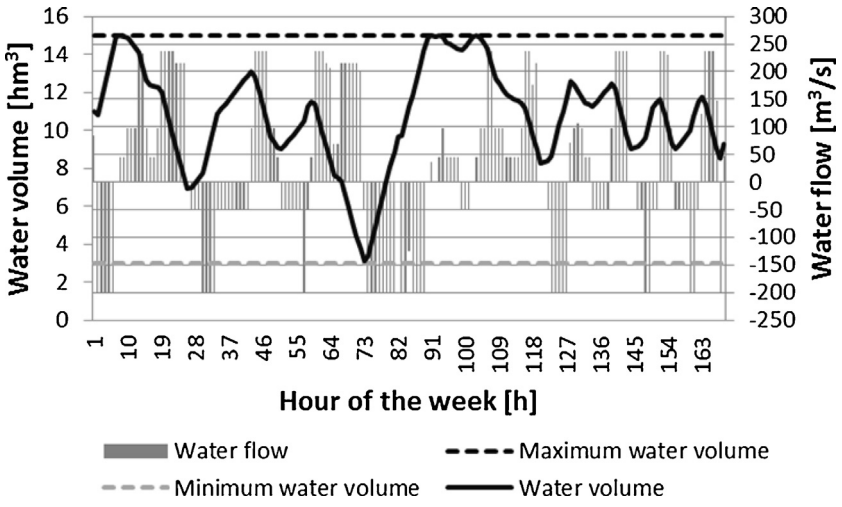


Fig. 12. Optimal water flow and volume of the pumped storage plant with Strategy 3 in scenario w_1.

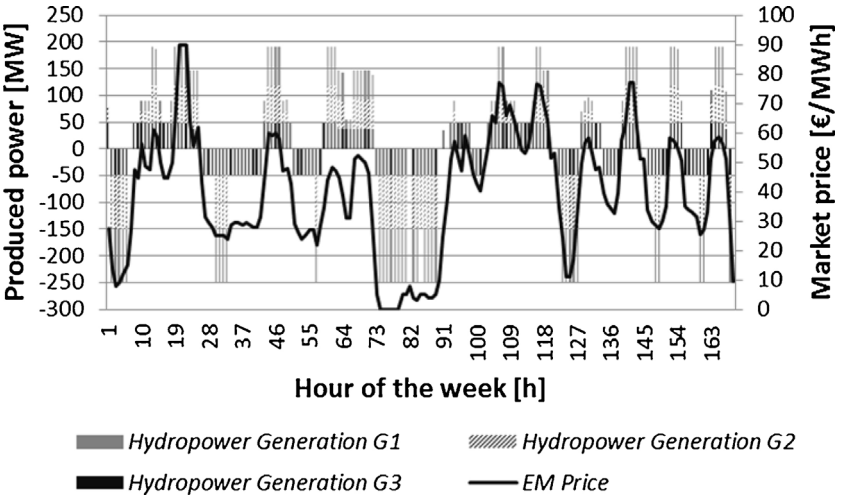


Fig. 13. Optimal power generation and consumption of the pumped storage plant with Strategy 3 in scenario w_1.

A sensitivity analysis is proposed in order to determine expected results in Strategy 3 for different values of $\mathcal{F}^{up}$ and $\mathcal{F}^{dw}$ parameters. Table 3 summarises main expected results for six cases. Each case is solved with a certain value of $\mathcal{F}^{up}$ and $\mathcal{F}^{dw}$ parameters. For instance, 25% case considers that the maximum percentage of the offered secondary regulation-up and down reserve requested by the TSO in real time is 25%, $\mathcal{F}^{up}$=$\mathcal{F}^{dw}$= 25%. The 2013 Mean case uses the mean values of the Spanish electricity system in 2013, which correspond to 31.9% for secondary regulation-up reserve ($\mathcal{F}^{up}$= 31.9%) and 23.7% for regulation-down reserve ($\mathcal{F}^{dw}$= 23.7%). Fig. 14 shows how the expected income and the expected energy and reserve offered in the corresponding markets vary with the assumed maximum percentage of secondary regulation reserve requested by the TSO in real time. The expected income, energy and reserve refer respectively, to the sum of the hourly average income, energy and reserve values across all scenarios weighted by their corresponding probabilities.

As can be seen in Fig. 14, the higher the assumed maximum usage of the regulation reserve, the lower the scheduled regulation reserve and therefore, the lower the expected income in the reserve market. By contrast, the higher the assumed maximum usage of the regulation reserve, the higher the energy and the expected income in the day-ahead market. This is a very rational result. A risky decision maker ($\mathcal{F}^{up}$=$\mathcal{F}^{dw}$= 0%) would offer more reserve regardless the uncertainty associated to the amount of reserve requested in real time by the TSO. In turn, a conservative decision maker ($\mathcal{F}^{up}$=$\mathcal{F}^{dw}$= 100%) would offer less reserve in order to be more protected against risk of water and storage unavailability in

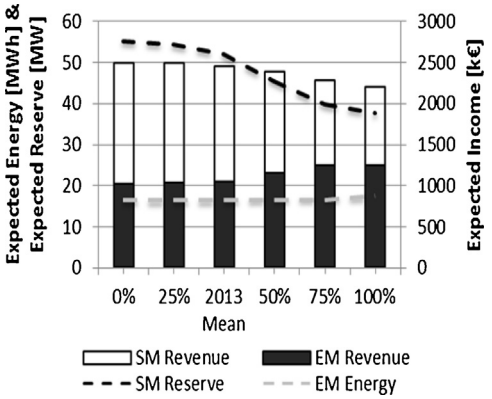


Fig. 14. Expected energy and reserve and expected income in the sensitivity analysis.

Table 3
Expected results in the sensitivity analysis.

	0%	25%	2013 Mean	50%	75%	100%
EM Energy (MWh)	16 497.25	16 474.61	16 541.3	16 934.87	17 057.83	17 626.99
SM Reserve (MW)	38 617.19	37 846.61	35 399.4	28 477.28	22 638.5	19 971.74
EM Revenue (€)	1 021 568	1 032 840	1 046 159	1 160 454	1 247 882	1 254 552
SM Revenue (€)	1 477 216	1 463 540	1 413 442	1 225 442	1 034 119	947 760
Total Revenue (€)	2 498 784	2 496 381	2 459 601	2 385 897	2 282 002	2 202 312
Income (€/MWh)	153.68	153.67	151.18	142.99	135.66	126.63

reservoirs with *bg* units due to the uncertainty in the real-time use of reserves, and would offer more energy in the day-ahead market whose uncertainty is considered in the scenario tree. As can be seen in Fig. 14 and Table 3, the more conservative the decision maker, the lower the expected total income and income per MWh; but the higher the reliability of the committed reserves.

4. Conclusion

This paper has presented a stochastic optimization model to find the optimal weekly schedule of a set of hydraulically coupled hydropower plants. The energy produced by every hydro unit, the regulation-up and down reserve that could be offered with protection against risk due to the uncertainty in the real-time use of reserves, and the detailed operation of the hydrosystem (reservoir levels at the end of each hour, water flows released through the turbines, and spillages) are profit-maximized in both the day-ahead electricity market and the secondary regulation reserve market, given the mentioned predominant role of hydro power in the provision of ancillary services. The future uncertainty about water inflows and market prices is considered through stochastic programming with a scenario tree. In hydropower scheduling problems participating in reserve markets, availability of secondary regulation energy is not always guaranteed since hydro operators do not usually consider the real time use of the committed reserves. By contrast, the proposed formulation protects a hydropower system against risk of water and storage unavailability due to the uncertainty in the real-time use of reserves. The case study shows that when a hydro producer participates in both the day-ahead electricity market and the secondary regulation reserve market, significantly higher expected income than when just selling electricity in the day-ahead electricity market can be obtained with the same consumed water volume. Additionally, the results show that it is possible to protect water and storage availability for the provision of the secondary regulation energy requested in real time, and to simultaneously increase significantly the income of the hydro system by participating in both markets, with respect to that obtained for participating only in the day-ahead electricity market.

5. Future work

Although regulation-up and down reserves are available with the solution of this model, uncertainty about reserve usage has not been taken into account explicitly. Introducing this uncertainty and evaluating its effects in the short-term hydropower scheduling problem is considered as future work. In addition to this, in certain electrical systems such as Spain, the secondary regulation-up and down energy is bought through auction mechanisms in liberalized markets. In the Spanish case, two extra sources of uncertainties (regulation-up and down prices) could be introduced in this model and they are also established as future work. Besides, the participation of pumped storage hydrounits in the secondary regulation reserve market is not taken into account in the presented model and it constitutes an interesting topic for future research.

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